

Math 151A: Applied Numerical Methods

Problem 1 (Newton’s method: how close is “sufficiently close?”) *Newton’s method can require the initial guess x_0 to be quite close to the root p to obtain convergence. Verify that this is the case for the root $p = \frac{\pi}{2}$ of*

$$f(x) = \cos x + \sin^2(50x).$$

Give a rough estimate of how small $|x_0 - p|$ should be in order to ensure convergence to p .

Hint. *First you must pick an appropriate interval I for a small neighborhood around p to guarantee convergence. Then on that interval determine the local bound*

$$|p - x_0| < \frac{1}{M}, \quad M = \frac{\max_{x \in I} |f''(x)|}{2 \min_{x \in I} |f'(x)|}.$$

For a rough estimate, you may use any suitable trigonometric inequalities to simplify M .

Solution. Let $p = \frac{\pi}{2}$ and $f(x) = \cos x + \sin^2(50x)$. Then

$$f'(x) = -\sin x + 50 \sin(100x), \quad f''(x) = -\cos x + 5000 \cos(100x).$$

Notice $|f''(x)| \leq 1 + 5000 = 5001$ for all x so that $M_2 := \max_{x \in I} |f''(x)| \leq 5001$.

Then for $f'(x) = -\sin(x) + 50 \sin(100x)$ notice that for $x = \frac{\pi}{2} + \delta$ that $f'(x) = -\cos(\delta) + 50 \sin(100\delta)$ and using an exact solver we find that the nearest zero is slightly larger than $\delta > 0.0002 > \frac{1}{5001}$. Therefore, we take $|p - x_0| < \frac{1}{5001}$

Problem 2 • *Show that the iteration equation for the Secant method can be written as*

$$p_n = \frac{f(p_{n-1})p_{n-2} - f(p_{n-2})p_{n-1}}{f(p_{n-1}) - f(p_{n-2})}.$$

- *In floating-point, discuss which form is less prone to cancellation when $p_{n-1} \approx p_{n-2}$ versus when $f(p_{n-1}) \approx f(p_{n-2})$.*

Solution. *Both forms suffer from cancellation in the denominator. But this form is more stable to cancellation in the numerator.*

Problem 3 (Newton with a huge initial guess) *Consider $f(x) = x^2 - 1$ and apply Newton’s method with $x_0 = 100,000$.*

1. *Derive the closed-form Newton update for this f .*

Solution.

$$x_{k+1} = x_k - \frac{x_k^2 - 1}{2x_k}$$

2. *Compute and tabulate the first 8–10 iterates.*

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# newtons step
def f(x):
    return x**2 - 1

def df(x):
    return 2*x

def newtons_step(x):
    return x - f(x) / df(x)

x0 = 100_000
for i in range(10):
    print(i,x0)
    x0 = newtons_step(x0)
print(i+1,x0)

```

```

0 100000
1 50000.000005
2 25000.000012500004
3 12500.000026250002
4 6250.000053125002
5 3125.0001065625
6 1562.5002132812444
7 781.2504266405786
8 390.6258533199398
9 195.31420665717374
10 97.65966330621758

```

Figure 1: Problem 3 Question 2.

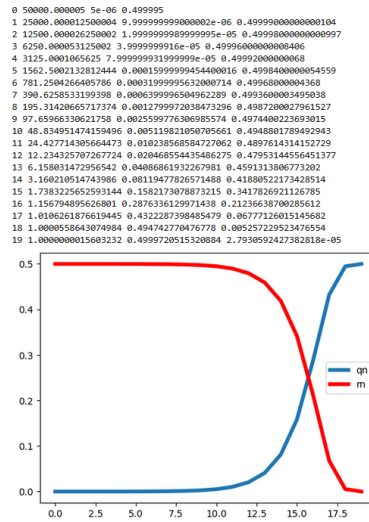


Figure 2: Problem 3 Question 3.

3. Let $x^* \in \{1, -1\}$ be the limit you observe. Estimate the order of convergence by

$$q_n = \frac{|x_{n+1} - x^*|}{|x_n - x^*|^2} \quad \text{and} \quad r_n = \frac{|x_{n+1} - x^*|}{|x_n - x^*|}.$$

What do these sequences suggest about the speed and order of convergence?

Solution. We see that as $q_n \rightarrow \frac{1}{2}$ and $a_n \rightarrow 0$ this indicates quadratic convergence with asymptotic error $\frac{1}{2}$.

4. Explain why this order is expected for Newton at a simple root.

Solution. For a simple root Newton's method converges quadratically at the rate of $\frac{f''(p)}{2f'(p)}$ which in this case is $1/2$.